The Distorted Time Framework

Local Rate as Primitive: Ontological Economy Without Global Structure

The rate of time as a physical primitive: gravity, inertia, and quantum mechanics

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Abstract

We take the rate of time as primitive: how fast a clock ticks, measured against its neighbours, is a physical fact, not a consequence of geometry. There is no global "now", and geometry follows from the rate.

One move, three consequences. Gravitational time dilation and quantum phase are *one* accumulated quantity: a redshift and an interference fringe measure the same clock. Gravitational and inertial mass are equal because there is only one rate. And local nows stitch into one cosmic time only if the cosmic flow has no twist; ours has almost none, which is a fact about how the universe began rather than anything the expansion produced, so cosmic time is an excellent approximation and not an exact fact.

Theories of this kind fail in two ways: those that make time global die on that global object, and those that keep it local must add one back to recover cosmology and entanglement. DTF needs neither. We start where the Wheeler–DeWitt equation does, with no time, and let time build from the local rates, so entanglement stays on that timeless layer rather than on a spatial slice. Realist quantum mechanics needs no preferred frame, and the correlations remain non-signalling.

This is an organizing-principle claim in the sense of Laughlin and Pines [1]: the content is the reorganization, not a new number. Where general relativity and quantum mechanics have been tested DTF agrees with both, and only the assumption count changes: the equivalence principle and cosmic simultaneity stop being separate postulates. It stakes one falsifiable claim, a horizonless, echo-free black-hole remnant replacing the singularity. Kill it and the reframe is dead.

1 Introduction — why this ontology

One commitment organizes this paper. The local rate at which a clock ticks is proper time, read from each point to its neighbours, and we treat it as a physical primitive rather than a gauge coordinate. Making it primitive does not relabel general relativity; it *forces* connections the separate tellings of gravity and quantum mechanics have no reason to draw. The sharpest: a particle's quantum phase accumulates at its internal frequency ω_C over proper time, and gravitational redshift is the variation of that same proper-time rate u, the lapse. Both run through one lapse,

$$\Phi = \omega_C \int u \, dt,$$

so time dilation and phase accumulation are two readings of a single accumulated Φ, though general relativity and quantum mechanics keep them in separate chapters. A reader who sees an accumulated phase and thinks of Fermat's principle is reading it correctly: making Φ stationary gives classical motion and summing over it gives quantum motion, which is the one variational rule used throughout (§2, §6). That mass *is* a frequency ω_C is de Broglie's clock [2], now measured [3] and formalised [4]; the new step is to make the *lapse* u primitive — *not* to call it ω_C, but to see that this one geometric rate both redshifts every clock and clocks every phase.

The moves, stated plainly. We do not ask the reader to uncover the commitments; here they are, each made good in the section named.

1. **Three rows, each emerging from the one before.**
   - *Row 0* — the timeless substrate ($\hat{H}\psi = 0$ [5], no external time [6]): the ledger of entanglement and conservation.
   - *Row 1* — the local clock-rate u emerges: time, energy, and the clock's tick-count N whose logarithm is the entropy; also gravity, inertia, guidance.
   - *Row 2* — space emerges as the rate's radiation: distance, the two gravitational-wave polarisations, $\hbar$, and coarse-grained *statistical* thermodynamics (temperature, heat, Hawking radiation) — emergent, never fundamental.

   (*Entropy* is not divided by this. There is one entropy, $S = \ln N$, the ordinary logarithm of the number of distinguishable states. The rows answer a prior question, *what makes states distinguishable*, and at Row 1 the answer is the clock. Heat is a Row-2 way of reaching a large N, not a second kind of entropy, so no Row-0/Row-1 derivation here may use a thermal quantity. Made good in §2 and Appendix G.)
2. **Energy is a frequency:** $E = \hbar\omega_C$ — mass is an internal frequency, which the lapse u modulates (u is the primitive, not ω_C) (§2).
3. **Space is time's radiation:** the radiating sector is the spatial metric's transverse-traceless part (two polarisations); the clock-rate itself does not radiate (§5).
4. **The now is local:** no global slice; a shared cosmic time emerges only as a limit. Entanglement is a Row-0 phenomenon that cannot signal (§6).

What these recover. From these blocks the standard results follow as *consequences, not inputs*: the equivalence principle (§4); Newtonian and weak-field gravity, and Kerr (§4); the two gravitational-wave modes with $\alpha_1 = \alpha_2 = 0$ (§5); Schrödinger's equation and the Born rule (§6); and the Tsirelson bound (§6). In every tested regime the framework is general relativity plus quantum mechanics.

What these unify, dissolve, and predict. Beyond recovery, the one primitive does three things GR and QM, told separately, do not:

- **Unify** — gravitational redshift = quantum phase (the shared Φ); gravity and inertia (one rate, §4); measurement and the black-hole core as one clock-definiteness surface (§6,§7); the galactic scale a_0 tied to the dark-energy horizon (§7).
- **Dissolve** — three questions stop being askable rather than being answered: why gravitational and inertial mass match (§4); why realist QM needs a preferred frame (it does not, since entanglement lives on the timeless Row 0, §6); and the black-hole information paradox (no singularity destroys information; it is held in Row 0, §7).
- **Predict** — a horizonless, echo-free black-hole remnant (§7): the one falsifiable owe; and, at plausibility level, the *value* $a_0 \approx cH_0/2\pi$ MOND leaves as a free constant (§7). The decaying frame-dragging residual (§2) is *not* offered as a test: it is unobservable by fifty orders of magnitude.

What we then show. The body makes each move good and consistent with existing mathematics and experiment, and is honest about the one place DTF commits to something new that could be wrong: a horizonless, echo-free black-hole remnant in place of the singularity (§7). **Kill it and the reframe is dead.** This is an organizing-principle claim in the sense of Laughlin and Pines [1] — its value is the reorganization, not a new number; the posit-count is quantified in Appendix F.

Why this set of moves, and not the neighbours'. The time-primitive neighbourhood is a graveyard where global-time-only and local-time-only theories all have well-known failure points. DTF solves for each of these issues.

Global time first. Neo-Lorentzian frames, dynamical clock fields (Einstein-æther, khronometric, Hořava [7, 8, 9]), varying-c, and preferred-foliation quantum theories [10] each die on a *global* time.

Local time first. Programmes that reject a background time recover the local fact only by *adding* a global object (a clock-subsystem correlation [11], modular flow [12], or a preferred foliation [13]). If they stay purely epistemic [14], they add no physics.

DTF escapes both. It is local, so it never owns the global object the first group dies on. And its rate is a real physical relation, not a bookkeeping convention: it is how fast this clock runs against its neighbour, which is what a redshift measurement reads. That is why it need not *add* a global structure the way the second group does: a shared cosmic time comes back as a limiting case, valid wherever the vorticity is negligible, as it is in our universe (§2).

Organization. §2 states the primitive and the removal of global structure; §4–§6 draw the equivalence, gravity, and quantum consequences; §7 collects the remaining owe alongside the agreement with GR and QM. The Technical Appendix holds the detailed recoveries and the economy ledger (Appendix F); scripts back every *verified* result (Code Availability).

2 The local rate and the removal of global structure

Works in: Row 0 → Row 1 (Wheeler–DeWitt sector → lapse sector)

Physics is usually built on space, time a coordinate. DTF wagers the opposite: the local rate at which clocks run (the ADM lapse) is a physical primitive, emerging from Row 0 (the Wheeler–DeWitt sector, $\hat{H}\psi = 0$, no time). The primitive is *relational*: how fast a clock ticks relative to its neighbours, chained point to point. Three consequences follow.

- **No global slice.** Each comparison needs only a local rest frame, and the choice is non-integrable, so no global u exists. That absence *is* the absence of a preferred foliation.
- **Gravity is GR's.** The rate does not propagate ($\pi_u \equiv 0$, standard ADM) and the dynamics stay Einstein's, so gravity, the two wave modes, and $\alpha_1 = \alpha_2 = 0$ are inherited. No gravitational-sector discriminator — the price of adding nothing.
- **The drive.** Covariantly u is the proper-time rate of the matter congruence n^μ, whose four-acceleration is the drive, $a^\mu = \nabla^\mu \ln u$ (§4).

The non-integrability is concrete: the vorticity of n^μ — the twist of the worldline congruence, the quantity frame-dragging measures — which by Frobenius forbids a global time function — and, for a given congruence, *is* the non-integrability measured as frame-dragging ($g_{0i} \neq 0$; GPB [15], LARES [16]).

So a *global* now is a theorem, not a posit: it exists *iff* the vorticity vanishes. One might expect the expansion to supply that limit, since

$$\omega \sim a^{-2}, \qquad \omega/H \to 0,$$

but the work it does is small: the smallness of cosmic vorticity is an *initial condition*, since single-field inflation excites no first-order vector modes. The expansion only mildly reinforces it; it does not drive the limit. Cosmic time is therefore inherited rather than manufactured, and standard cosmology is recovered as a gauge, not contradicted (`DTF_localnow_integrability.py`). The same division of labour as the arrow (§2 above): the framework supplies the boundary condition and inherits the dynamics.

The numbers behind that sentence. Three facts settle how much the expansion does and whether the residual can be measured.

- **The decay law is exact.** $\partial_\tau(Sa^2) = 0$ for the vector metric perturbation, so $\omega \sim a^{-2}$.
- **The expansion does little of the work.** In the radiation era $H \propto a^{-2}$ as well, so ω/H is *frozen*. The approach to integrability begins only at matter–radiation equality, and has achieved a factor of 74 since.
- **The residual is unobservable.** At horizon scale $\Omega \sim H_0 S/2$ gives $\lesssim 10^{-11}$ mas/yr, against Gravity Probe B's 7.4 mas/yr; the tightest bound on a homogeneous vector mode, $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ [17], sits fifty orders above any realistic seed.

The residual is therefore **not a test**. DTF inherits general relativity's vector sector equation for equation, so this is a consistency check the reframe passes rather than a place it sticks its neck out (`DTF_vorticity_residual.py`).

Two boundaries: the holographic IR cutoff is the causal (future-event) horizon, not the Hubble sphere, so $w \to -1$ is not forced (Hsu–Li [18, 19]); and the rate is *not* prior to the metric — clock-vs-neighbour is co-equal with the causal order (c) it is defined against, which is what *relational* means.

The primitive is the local relational rate u — clock-vs-neighbour, no global "now"; nothing is added beyond taking it as physical.

$$S = -mc^2 \int d\tau$$

[*known: a worldline accumulates proper time (particle sector, Row 1)*]

Known

[DTF input:] the primitive is the local relational rate — clock vs neighbour, no global slice; the lapse only as read in a chosen coordinate ($\pi_u \equiv 0$, standard ADM); not an independent field

DTF input

A single global rate field would need a slice; DTF asks only for *local* comparisons, which need only a local rest frame, not a global slice — and that local choice is non-integrable, so no global u exists (the non-integrability *is* the absence of a preferred frame). Path-dependent at a distance, like proper time. The field equations remain Einstein's, so DOF= 2 and no scalar mode — inherited from GR.

Derivation

no structure added to GR; $\pi_u \equiv 0$, DOF $= 2$, no preferred frame

DTF result

the primitive is a local relational rate; gravity is GR, frame-neutral

Takeaway

Status. Nothing added — contrast shape dynamics [13] and Hořava [9], which *do* add a preferred foliation and (in Hořava) a propagating scalar; DTF adds neither.

The engine: energy is a frequency, the lapse is the rate

With the primitive fixed we read dynamics off, not add them. Two distinct "rates" meet here, and keeping them apart is the whole point:

- **The Compton frequency** $\omega_C = mc^2/\hbar$ — the particle's internal clock, *per-species* and dimensionful (Hz). That energy is this frequency ($E = \hbar\omega_C$, with $E = mc^2$) is de Broglie's clock [2], measured in electron channeling [20] and as a Compton-frequency clock [3], and formalised by Dolce [4] — established prior art, not ours.
- **The lapse** $u = d\tau/dt$ — the local proper-time rate, *per-location*, geometric and dimensionless (the same for every species). This is the primitive.

These are *not* equal. The new step, which de Broglie and Dolce do not take, is to make u the primitive and to notice that it does two jobs GR and QM keep apart: it *redshifts* the clock (gravitational time dilation is the variation of u) and it *clocks* the phase, $\Phi = \omega_C \int u\, dt$. So redshift and phase are two readings of the one lapse (the shared Φ), because both run through u, not because ω_C is u. Whether the local u's admit a global time function is then a separate Frobenius question (§2), answered only in the vorticity-free limit, so cosmic time is emergent, not posited.

One drive then follows, matter falling toward slower clocks, under one rule: Fermat's principle on Φ, read at two sharpnesses (stationary = classical mechanics, §4; full sum = quantum mechanics, §6).

Everything below is forced *by the interval; energy is not a second ingredient.*

$$S = -mc^2 \int d\tau = -mc^2 \int \sqrt{u^2 - v^2/c^2}\, dt$$

[*known: the full interval (not the rest form)*]

Known

$E = -\partial_t S = \gamma\,\hbar\omega_C \to \hbar\omega_C$ at rest. In the weak ($u = 1 + \phi/c^2$), slow ($v \ll c$) limit $\mathcal{L} = -mc^2 u + \frac{1}{2}mv^2$, whose Euler–Lagrange equation is $\ddot{\mathbf{x}} = -c^2\nabla u = -\nabla\phi$. (The rest form $\int u\,dt$ alone gives $\nabla u = 0$; the heading term is required.)

Derivation

$$E = \hbar\omega_C, \qquad \ddot{\mathbf{x}} = -c^2\nabla u$$

DTF result

energy is *the internal frequency ω_C; the drive falls down the gradient of the lapse u*

Takeaway

One drive, one rule, two sharpnesses. The drive is $\ddot{\mathbf{x}} = -c^2\nabla u$; the rule is Fermat on the accumulated clock reading $\Phi = S/\hbar = -\omega_C\int d\tau = -\omega_C\int\sqrt{u^2 - v^2/c^2}\,dt$ — the *one* variational object used throughout (its rest reduction $\int u\,dt$ drops the heading v^2 term and gives $\nabla u = 0$, so the full interval is required; §6 varies this same Φ). Stationary path ⇒ gravity and inertia (§4); full sum ⇒ quantum mechanics (§6). The three constants are one tick seen three ways: c (period → length, Row 1→2), $mc^2 = \hbar\omega_C$ (rate in energy units, Row 1), $\hbar$ (the localisation tax $\lambda_C = \hbar/mc$, Row 2).

Status. Same Φ throughout (`DTF_phase_unification.py`). Two rates, not conflated: ω_C is the particle's Compton tick, u the gravitational rate that redshifts it ($d\tau = u\,dt$ at rest) — so "one primitive" means one *kind* of rate, not one number (masses stay inputs, Appendix E).

3 Entropy and the arrow of time

Works in: Row 0 → Row 1 (the timeless state → the clock's tick-count)

Entropy here means the ordinary microstate count,

$$S = \ln N,$$

the logarithm of the number of distinguishable states a system has reached, and the paper uses no other. That the thermal entropy of a bath follows from this count is textbook and is not re-derived; the question taken up here is the one before it. The rate primitive answers a question statistical mechanics takes as already settled: *what makes states distinguishable at all.* Temperature cannot be that answer, because it is *defined* from the entropy ($1/T = \partial S/\partial E$) and so presupposes the count. The primitive can be: N is the number of ticks the clock has resolved since the Row-0 state, and the same N is read three ways,

$$E = N\hbar\omega \ \text{(linearly)}, \qquad \Phi = 2\pi N \ \text{(as phase)}, \qquad S = \ln N \ \text{(as a log)},$$

with temperature nowhere among them.

- **The zero is definitional.** Row 0 is one synced state: $N = 1$, so $S = 0$ exactly. The "past hypothesis" [21] is the low-entropy start that a time-symmetric dynamics must be handed. It is not postulated here; it is what *being* Row 0 means.

- **Ticks are what become distinguishable.** A spread σ_u in the local rates desynchronises phases, resolving $N \simeq \omega\,\sigma_u t$ states, so $S = \ln N$ grows as $\ln t$. The simulation checks the *bookkeeping*, not the arrow: the fit $S = a \ln t + b$ returns $a = 0.982$ against the predicted 1, and doubling *either* ω or σ_u shifts S by 0.693 against $\ln 2 = 0.6931$ — confirming that both enter multiplicatively *inside* N and never additively in S. Since $N \propto \omega\sigma_u t$ was put in, this is a consistency check on a definition, not evidence for it.

- **Growth is entanglement, not loss.** Row 0 stays globally pure ($S_{\text{global}} = 0$ at all times); what rises is a subsystem's share, linearly in the tick count (slope 0.66 in a transverse-field Ising quench). The arrow is the Row-0 ledger distributing correlations, which is also why nothing is destroyed at a black-hole core (§7). Consistent with experiment: in an isolated six-atom lattice the global state stays pure while subsystems reach the values a thermometer would report [22] — an illustration of the reading, not a test of it.

- **Counting before heating is standard.** Landauer's erasure cost $k_B T \ln 2$ already separates the count $\ln 2$ from its price $k_B T$ [23, 24]; a count survives at $T \to 0$ [25]; the Kolmogorov–Sinai rate counts *per unit time* with no temperature in it [26]. None of this is new. The one new step is to identify N with the clock.

- **Where the arrow comes from, and where it does not.** The division of labour is worth stating exactly, because only half of it is ours. The desync law is *time-symmetric*: run $\phi_i = \omega u_i t$ backwards and the clocks resynchronise. With commensurate rates it recurs (Poincaré). So the growth of N is ordinary mixing, inherited from statistical mechanics and not derived here; what breaks the symmetry is the $N = 1$ boundary condition, and *that* is what the rows supply. DTF contributes the piece Boltzmann must assume and inherits the piece he already had. The mechanism that grows σ_u from the synced state is gravitational clumping, argued from the sign of the instability rather than simulated (O5).

- $\ln N$ **is a ceiling, not the value.** $S = \ln N$ is the equal-weight (microcanonical) reading; the general entropy is $-\sum p \ln p$, which is smaller whenever the resolved states are unevenly weighted. For a Gaussian spread in u the gap is an additive constant of order $\ln\sqrt{2\pi e}$ (measured: 0.70–0.79 nats, slowly varying), so the growth law is unaffected and every rate statement above stands; what is not licensed is reading $\ln N$ as the entropy itself. Write $S = \ln N + O(1)$.

The thermodynamic firewall. A bath is one way of resolving many states — put a system in contact with one and $N \sim k_B T/\hbar\omega$, after which the same $S = \ln(eN)$ returns the textbook number. It is a *way of reaching* a large count, not a different entropy, and it is available only once Row 2 exists. Hence: **no Row-0 or Row-1 derivation in this paper may use a thermal quantity.** The rule has teeth: it excludes an a_0 read off an Unruh temperature, which is why Appendix B takes the geometric route, and it excludes a temperature-triggered de-emergence at the black-hole core, which is why §7 uses a Row-native $\delta u/u$ trigger. It also separates this from Jacobson's and Verlinde's entropic gravity [27, 28] and from thermal time [12], which run a Row-2 temperature down into Row 0/1.

The firewall pays a second debt. Penrose's objection [29] — the early universe is hot, so how can it be the low-entropy start? — is a question about *shares of one count*. Measured budgets [30] put

the black-hole share of the observable universe at $3 \times 10^{104} k_B$ against the microwave background's $\sim 10^{88} k_B$, with the cosmic event horizon at $2.6 \times 10^{122} k_B$: radiation holds a negligible fraction of N, and the dominant share — the one carried by clumping, by departures of u from uniformity — is near its minimum when u is uniform. A hot smooth start is therefore a *small* N, and the Weyl-curvature hypothesis becomes a statement about which share dominates rather than an added postulate. Two honest limits: how to count the clumping share has no agreed formula (black-hole entropy is the proxy), and the dynamics that grows σ_u, gravitational clumping, is argued from the sign of the instability rather than simulated (O5).

Status. Verified: `DTF_native_arrow.py` (a=0.982, $\ln 2$ shifts), `DTF_entropy_clockcycle.py` ($E{=}N\hbar\omega$ and $S{=}\ln N$ from one N), `DTF_entropy_is_entanglement.py` ($S_{\text{global}}{=}0$), `DTF_arrow_grav_thermal.py` (shares of the count). No temperature enters any Row-1 result.

4 Equivalence and inertia as one rate

Works in: Row 1 (the well in u; near-zone at 1PN, Kerr inherited, in Row 2)

Gravity and inertia are one rate from two sides. Matter carries energy, energy slows clocks, a slow clock is a dent in u, and a free body falls down it: Newtonian gravity. To accelerate is to re-rate the clock, which costs exactly the kinetic energy. One ω_C both sources the well and resists re-rating, so gravitational and inertial mass are a single identity.

The primitive removes a coincidence. GR *assumes* $m_{\text{grav}} = m_{\text{inertial}}$; read from one rate there are not two masses to equate — the equivalence principle is not explained but *dissolved*, a coincidence that never had two sides, the strongest case in the ledger. The same identity fixes the coefficient: re-rating costs $mc^2(\gamma - 1)$ with $mc^2 = \hbar\omega_C$ definitional (§2), so no self-energy mechanism is owed.

Energy slows clocks; the drive of §2 does the rest.

$$\ddot{\mathbf{x}} = -\nabla\phi$$

[*known: Newtonian free fall*]

Known

In the weak-field limit the rate obeys $\nabla^2 u = \frac{4\pi G}{c^2}(\rho + 3p/c^2)$ — the Tolman active mass, exactly the combination that gravitates (reducing to ρ for dust) — the ordinary GR weak-field equation, no slicing invoked. A moving clock has $E = \gamma mc^2$, so setting it in motion costs $\Delta E = mc^2(\gamma - 1) \to \frac{1}{2}mv^2$.

Derivation

$$\nabla^2 u = \frac{4\pi G}{c^2}\left(\rho + \tfrac{3p}{c^2}\right), \qquad \Delta E = mc^2(\gamma - 1)$$

DTF result

gravity is the well in u (GR weak field); inertia is the cost to re-rate the same ω_C

Takeaway

(The source is the *active* gravitational mass $\rho + 3p/c^2$, not ρ alone: pressure carries energy and so also slows clocks — the Komar/Tolman combination, unchanged from GR.)

Status. Near-zone at 1PN via the conformally-flat scheme [31] (Kerr has no such slice [32]); all-orders inheritance via the Deser spin-2 bootstrap [33], whose equations are Einstein's, so vacuum solutions (Kerr included) are inherited, not reconstructed (`DTF_kerr_deser_closure.py`). No tension with §5: Deser is the *classical* spin-2 completion, not a quantized mediator. The rotating interior is GR's open problem, inherited unchanged.

5 Gravity without a force-carrier

Works in: Row 1 → Row 2 (radiative sector of the spatial metric)

Gravity is the gradient of the primitive (§4), so the attraction is carried by nothing — no field whose quantum mediates it. The spatial metric's radiation propagates instead, and structure fixes its content: when the metric ripples it carries the two transverse-traceless polarisations of a gravitational wave. The constrained lapse (§2) does not ripple — no scalar "breathing" mode, no preferred-frame effect [34], and (unlike Hořava [9]) no leftover scalar. A breathing-mode detection would falsify DTF and GR alike: shared falsifiability, not a discriminator.

The primitive buys no force-carrier. In the field picture a spin-2 graviton mediates the pull by exchange; here the pull is the u-gradient and the two modes are space's *radiation*, not a mediator. There is no spin-2 force field to quantise — this removes the force-carrier, not the radiative quanta (the two modes are gravitational-wave quanta, not carriers of the static pull).

The spatial metric is linearised about flat space, $\gamma_{ij} = \delta_{ij} + h_{ij}$.

$$\gamma_{ij} = \delta_{ij} + h_{ij}$$

[*known: extents assembled into a spatial metric (Row 2)*]

Known

[DTF input:] radiation is in the spatial sector; the lapse is constrained ($\pi_u \equiv 0$), so it does not propagate

DTF input

$$h_{ij}^{TT}: \ 2 \text{ tensor pols}; \quad \pi_u \equiv 0 \Rightarrow \alpha_1 = \alpha_2 = 0$$

`DTF_radiative_content` → no breathing mode; $\Delta = 0$ two ways

DTF result — verified

Status. Two tensor modes and $\alpha_1 = \alpha_2 = 0$ *identical to GR* — shared falsifiability, no discriminator. The ADM phase-space count and the explicit $\alpha_1 = 0$ at 1PN are in Appendix A (`DTF_alpha1_explicit_1PN.py`).

6 The quantum rate field

Works in: Row 1 (single particle); Row 0 (entanglement); the tax is Row 2 ($\hbar$)

This section adds no equation to quantum mechanics: it is a *compatibility* reading, importing Madelung, Fisher-uniqueness, Bohmian guidance, and Born-as-typicality wholesale and asking only what the rate primitive reidentifies. Its one net-new element is that the phase is a clock reading, which gives Wallstrom's condition a physical reason (now shown dynamically stable, §6); everything else is standard, relabelled.

The Fermat principle at its *other* sharpness is quantum mechanics, and the wave is not abstract. Write

$$\psi = \sqrt{\rho}\, e^{i\Phi}, \qquad \Phi = S/\hbar.$$

Both symbols already have meanings from §2:

- Φ is the accumulated clock reading.
- ρ is the density of definite clock-rate; a particle is a localized excitation of it above the Row-0 floor.

So the wavefunction *is* the rate field rather than an abstraction over it.

One honest cost. That literal reading holds for one particle in ordinary space. For N particles the guiding wave lives on the $3N$-dimensional configuration space, so the ontology carries $O(N^2)$ correlational structure where a naive space field would carry $O(N)$. Every realist quantum theory pays this, and it is why entanglement is booked on Row 0 rather than as a field in space.

Substituted into Schrödinger's equation, the polar form splits *exactly* [35]:

$$\underbrace{\partial_t \rho + \nabla\cdot\Big(\rho\,\frac{\nabla S}{m}\Big) = 0}_{\text{continuity}} \qquad \underbrace{\partial_t S + \frac{(\nabla S)^2}{2m} + V + Q = 0}_{\text{the clock-action, plus one term}}$$

and that one extra term is the localization tax,

$$Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} = \frac{\delta}{\delta\rho}\int \frac{\hbar^2}{8m}\frac{(\nabla\rho)^2}{\rho}\,d^3x,$$

the variation of the Fisher functional [36, 37].

DTF adds no term. It reads the two already there, and three results follow.

- **The tax is forced, not chosen.** Among local terms with at most two derivatives, the Fisher form is the *unique* one that keeps the ψ-equation linear (`DTF_o9_fisher_unique.py`). Why quantum mechanics is linear is not addressed.
- **Born is not a separate axiom.** The current $j = \rho\nabla S/m$ makes $|\psi|^2$ equivariant, so Born frequencies are a typicality assumption [38, 39] rather than a theorem, and objects stay definite under $v = \nabla S/m$ [40]. One residual remains: which outcome a lone detection selects (O1).

- **Discreteness is topological.** A clock reading is S^1-valued, so ψ is single-valued with integer circulation around nodes. This gives Wallstrom's condition [41] a physical reason, and the winding stays exactly integer through moving nodes (`DTF_winding_dynamics.py`). (Fermionic $\psi \to -\psi$ is spin structure, not a counterexample; single-valuedness is an input here.)

Entanglement. It needs no cosmic "now". A joint state's correlations live in the *timeless* Row-0 ledger, not on a slice: each outcome is definite *locally*, and the correlation appears only when results are brought together. Nothing orders spacelike-separated outcomes, so the account is frame-neutral (contrast relativistic pilot-wave theories [10, 42], which must postulate a foliation). The ledger is timeless rather than a past hidden variable, so it evades Bell's factorisation and is compatible with the quantum bound. The CHSH operator obeys

$$C^2 = 4I - [A, A'][B, B'], \qquad \text{so} \quad \|C\| \le 2\sqrt{2}$$

(the standard Tsirelson bound, which QM saturates; the classical 2 is the commuting case; `DTF_o8_tsirelson.py`). This is the recognised locally-mediated/atemporal class [43], Row 0 the timeless bookkeeper. What remains: a dynamical ledger, and no-signalling, which the row ordering *motivates* without yet deriving.

No-signalling, and how far the rows account for it. Read the pair as *one* Row-0 object Ω with two handles rather than as two systems, and let π be the projection whose *output* is space:

$$P(\alpha, \beta \mid x, y) = \big|\langle \alpha\beta \mid \pi(x, y)\, \Omega\rangle\big|^2, \qquad \sum_\beta |\beta\rangle\langle\beta| = \nVdash.$$

The second equation is completeness of the reads on each handle, which here is Born as the conserved rate-density. It erases any handle-local far operation from the near marginal, so the far setting y could reach it only by riding an operator that couples the two handles. Such an operator relates handle *locations*, and locations are downstream of π. Hence

$$\partial_y \sum_\beta P(\alpha, \beta \mid x, y) = 0$$

while the joint keeps both settings and Bell remains violable.

- **What this does not do.** It does not *derive* the restriction. The same computation returns zero for *every* link value, including the unentangled $\theta = 0$, so no property of the Row-0 object is doing work. The one active ingredient is the product form of π, the handle tensor structure itself, and writing Ω in a product space already presupposes the separability whose consequence is at issue.

- **What it does do.** It gives a *reason to expect* locality of reads — space is an output, so Row 1 offers only handle-local ones — where the atemporal accounts simply posit it.

The residue is sharp and is listed as O3.

Status. Verified: `DTF_no_signalling_projection.py`.

The wave is the rate field; the polar form splits Schrödinger into conservation plus the tax.

$$\psi = \sqrt{\rho}\, e^{iS/\hbar}$$

[*known: the rate field: ρ its density, $S/\hbar$ the clock reading (§2)*]

Known

[DTF input:] the rate-stuff is conserved (ρ obeys continuity) and localizing it costs the tax $\int \frac{\hbar^2}{8m}(\nabla\rho)^2/\rho$

DTF input

Continuity plus §2's classical Hamilton–Jacobi equation plus the tax's variation $Q = -\frac{\hbar^2}{2m}\nabla^2\sqrt{\rho}/\sqrt{\rho}$ recombine *exactly* into one complex equation. The current $j = \rho\nabla S/m$ makes $|\psi|^2$ the unique equivariant measure; detection is all-or-nothing.

Derivation

$$i\hbar\,\partial_t\psi = -\tfrac{\hbar^2}{2m}\nabla^2\psi + V\psi, \quad Q = \tfrac{\delta}{\delta\rho}\int \tfrac{\hbar^2}{8m}\tfrac{(\nabla\rho)^2}{\rho}$$

`DTF_localization_tax` → split exact; Q is the tax

DTF result — verified

A declined discriminator. A 2025 measurement [44], building on earlier density-gradient work [45], seemed to hand DTF one, and we declined it. Where $\psi \sim e^{-\kappa x}$ is real the current vanishes ($\nabla S \to 0$) and the beable is at rest, yet a nonzero transit speed is measured — which the tax reproduces as $v_{\rm op} = \sqrt{2|Q|/m} = \hbar\kappa/m$ (`DTF_evanescent_speed.py`; Appendix C). We built nothing on it: the value is dimensionally automatic (as with a_0), the operational speed is not a beable velocity, and a directed amplitude term would break Born equivariance; the underlying challenge to Bohmian mechanics has since been rebutted [46, 47] (authors maintaining their position [48]). DTF keeps $v = \nabla S/m$; whether the tax could nonetheless act as a second channel is settled next. *Could the localization tax carry a* second *beable channel — a velocity built from the amplitude rather than the phase, alive exactly where* $\nabla S \to 0$*? The general condition settles it, and the special case that motivated the question is one line of it.*

$$v = \frac{\nabla S}{m}, \qquad j = \rho\, v, \qquad \partial_t\rho + \nabla\cdot j = 0$$

[*known: the phase channel: one beable current,* $|\psi|^2$ *equivariant*]

Known

[DTF input:] a second channel adds *any* current: $v \to v + w/\rho$

DTF input

Equivariance, meaning $\rho = |\psi|^2$ is preserved at *all* times, holds iff the addition is divergence-free, $\nabla{\cdot}w = 0$ (Deotto–Ghirardi [49]). That single condition exhausts the possibilities:

$$\nabla{\cdot}w = 0 \;\Rightarrow\; \text{every single-time statistic unchanged;} \qquad \nabla{\cdot}w \neq 0 \;\Rightarrow\; \big\|\rho - |\psi|^2\big\|_1 \simeq 0.7\,\varepsilon,$$

the latter measured for an admixture ε in a free packet, and bounded by the precision to which Born is confirmed. The one compensated case — restoring equivariance with noise of matching strength — is Nelson's stochastic mechanics, whose diffusion cancels the drift's divergence identically,

$$\partial_t\rho = -\nabla{\cdot}\big(\rho[v + D\nabla\ln\rho]\big) + D\nabla^2\rho = -\nabla{\cdot}(\rho v) \underbrace{- D\nabla^2\rho + D\nabla^2\rho}_{=0},$$

returning continuity exactly, and Nelson mechanics is empirically standard quantum mechanics. *Illustration.* An amplitude-directed law $w = f(\rho)\nabla\rho$ cannot even reach the first branch: $\nabla{\cdot}w = f'|\nabla\rho|^2 + f\nabla^2\rho$, and at a local extremum of ρ the first term vanishes while $\nabla^2\rho \neq 0$, forcing $f = 0$ there; an extremum sits at any density value, so $f \equiv 0$.

Derivation

deterministic ⇒ off-Born; compensated ⇒ QM; $\nabla{\cdot}w = 0$ ⇒ single-time invisible

DTF result

no beable channel discriminates on single-time statistics

Takeaway

The trichotomy is exhaustive for a beable with a current and optional noise, and since Schrödinger's equation is untouched (§6) that is the only place a discriminator could hide. *Does the tax source a surviving channel?* The divergence-free additions $w = \nabla\times\chi$ are the Deotto–Ghirardi non-uniqueness, and they matter here only if the localization tax fixes some $\chi \neq 0$. It does not, for an elementary reason: the current is the quantity conjugate to the phase,

$$j = -\frac{\delta\mathcal{A}}{\delta(\nabla S)} = \frac{\rho\nabla S}{m}, \qquad \mathcal{A} = \int\Big[\rho\,\partial_t S + \frac{\rho(\nabla S)^2}{2m} + \rho V + \underbrace{\frac{\hbar^2}{8m}\frac{(\nabla\rho)^2}{\rho}}_{\text{the tax}}\Big],$$

and *the tax contains no* S, so it contributes exactly zero to j. Its entire dynamical role is the scalar $\delta(\text{tax})/\delta\rho = Q$ in the phase equation — and a scalar potential shifts S, whose gradient is already the current. Hence $\chi = 0$: a divergence-free channel would have to be postulated by hand, and this framework postulates none.

The ambiguity is removed, not inherited. Deriving the current from an action rather than positing a guidance law is what fixes χ: theories that postulate guidance independently inherit the Deotto–Ghirardi freedom, and a chosen $\chi \neq 0$ does shift finite-aperture detection rates (a -4.8% example, resolved at our simulation's sample size — a statement about numerical resolution for that χ, not about detectability). With $\chi = 0$ forced, none of it is available here. **DTF is identical to quantum mechanics not only in single-time statistics but in arrival statistics too, and its whole empirical exposure is gravitational (§7).**

Status. Verified: `DTF_o2_amplitude_channel.py` (trichotomy; 0.7ε rate; Fokker–Planck cancellation exact), `DTF_o2_tax_sources_chi.py` ($\chi = 0$: the tax contributes $(0, 0)$ to j), `DTF_o2_arrival_time.py` (complete-plane flux null; aperture shift for $\chi \neq 0$), `DTF_o2_deotto_ghirardi.py`.

Compatibility: electromagnetism. Since only phase *differences* are physical, the origin has no global referent — fixing one would be exactly the global structure §2 forgoes — so it resets locally, and the standard $U(1)$ construction applies verbatim [50, 51]: A_μ is a clock-phase connection, $F_{\mu\nu}$ its holonomy. Minimal coupling adds the Lorentz force to the drive,

$$m\ddot{\mathbf{x}} = -mc^2\nabla u + q(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

and the lowest-order gauge-invariant dynamics is Maxwell (`DTF_em_from_phase.py`; Appendix D). The gauge *principle* is then not an added posit but the no-global-structure core read on the phase, and the surviving group is the frame-*orthogonal* $U(1)$ — gauging the frame's own internal rotations yields mass from the metric's vacuum, not a photon (Appendix D). It remains a *consistency check, not a derivation*: DTF fixes neither α nor the charge spectrum and makes no non-abelian claim.

7 Consistency and limits

The core claim is complete with the preceding sections. This section states plainly what the reframe owes and where it agrees with existing physics — in that order, because the owes are the point. Where general relativity and quantum mechanics have been tested, the framework agrees with them (it adds no equation there); that agreement is a consequence of the same locality that forces the owes, not a hedge against them.

That identity is *not* a reinterpretation dressed as physics, for three reasons.

- **Roles are removed, not constants.** The equivalence principle, the energy–frequency bridge law, and a preferred frame for realist quantum mechanics stop being separate assumptions and are read from one primitive (the ledger of §1). The dimensionful inputs ($c, G, \hbar, H_0$, the mass spectrum) are unchanged (Appendix E).
- **A structure is forbidden.** The non-propagating clock ($\pi_u \equiv 0$) is not a DTF stipulation but the ADM lapse's inherited status (Appendix A), a constraint any free retelling would have to respect.
- **A scale gets a reason.** A global-now general relativity supplies none for MOND; a local-now one does, because a local now has only a finite causal horizon and that horizon *is* an acceleration of order cH_0 (§7(ii), Appendix B). This tethers a_0; it does not derive the force law.

Beyond the tested regimes the falsifiable content is narrow and labelled as such: the black-hole core (conjecture) and the MOND scale (plausibility). Within quantum mechanics no channel survives the closure of §6, so no laboratory test can discriminate.

Result	**Equation**	**Status**
Gravity, GR limit	$\nabla^2 u = \frac{4\pi G}{c^2}(\rho + 3p/c^2)$; Kerr	identical to GR
Two tensor GW pols; $\alpha_1{=}\alpha_2{=}0$	h_{ij}^{TT}	identical to GR (shared falsifiability)
Inertia / equivalence	$\Delta E = mc^2(\gamma - 1)$	identical to GR
Quantum mechanics, Born	Schrödinger = clock-action + tax; $\|\psi\|^2$	identical to QM (Born: typicality)
Evanescent transit speed	$v_{\rm op} = \sqrt{2\|Q\|/m}$ (operational)	order matches [44]; not a Bohm test [46]
Black-hole core	de-emergence, remnant	conjecture
Dark sector / MOND	μ, $a_0 = cH_0/2\pi$	explains a_0 (plausibility)

Three applications push u to its extremes. The economy claim of §1–§6 rests on none of them — local-now and the Bell account stand if all three fell — but one, the MOND scale, is where the ontology makes contact with a live anomaly and so is the paper's so-what against a rival algorithm.

(i) The black-hole core — de-emergence, not singularity. The *static* lapse vanishes at the horizon, but a regular infaller crosses it with nonzero rate; time does not de-emerge there. Inside, read the crossing through a geodesic comoving clock (Oppenheimer–Snyder dust [52], horizon-regular): curvature tracks density, so $L_{\rm curv} \sim \ell_P$ becomes $\rho \sim \rho_P$, reached at *finite* proper time before the classical crunch,

$$\tau_{\rm rem} = \frac{1}{\sqrt{6\pi G \rho_P}} = \frac{t_P}{\sqrt{6\pi}} \approx 0.23\, t_P.$$

The clock reaches $\delta u/u \sim 1$ at *finite* scale factor — never $a = 0$. What de-emerges is its *coherence* ($N = L_{\rm curv}/\ell_P$ Planck ticks per curvature time; the crossing $N = 1$ is coefficient-free for any $p \in [\frac{1}{2}, 1]$), not the proper time — and Row-natively, *not* via temperature (a Row-2 concept whose periodicity-in-time cannot govern the failure of time without circularity). The surface sits parametrically inside the horizon: a fat, finite, Planck-density core, not a point, robust to inhomogeneity (a shell-by-shell front, `DTF_LTB_collapse.py`).

Remnant, not bounce. Evaporation halts at a horizonless relic,

$$M_{\rm rem} = m_P/\sqrt{8},$$

and the de-emerged energy stays in Row 0 rather than bouncing: there is no Row-1 time at the core to reverse in (collapse *exits* Row 1), the energy still gravitates by Einstein's equations, and a static core has no re-emergence driver. Hence a stable relic and *no* post-ringdown gravitational-wave echoes (the horizon is regular) — a falsifiable split from Planck-star phenomenology. (Whether such relics contribute to cosmological dark matter is a separate open question; DTF's dark sector is the MOND shadow below, *not* particle relics.)

This is *one definiteness surface*, shared with §6: a clock *becoming* definite (Row 0→Row 1) is quantum measurement; a clock *de-emerging* (Row 1→Row 0) is the core; the timeless Row-0 entanglement ledger sits on the far side of both. Only the $O(1)$ coefficient of $\delta u/u \sim (\ell_P/L)^p$ is open (O2; `DTF_singularity_deemergence.py`, `DTF_singularity_observable.py`).

The information paradox dissolves the same way. There is no singularity to destroy information: infalling matter de-emerges from Row 1 to the *timeless* Row-0 ledger — the same ledger that already holds entanglement correlations (§6). The correlations are not erased, only removed from Row-1 time; Hawking radiation (a Row-2, statistical property) carries off energy, while the information sits in Row 0

and in the stable horizonless remnant. It appears thermal only because its entanglement partner de-emerged to Row 0 rather than being destroyed — trace out that partner and one gets thermality, keep it and the total state is pure. This evades the usual species/entropy objection to remnants: the information is *not* in the remnant's local degrees of freedom (whose capacity is set by its Planck mass) but in Row 0, whose capacity is not. Loss of unitarity was an artifact of assuming a singular endpoint in Row-1 time; with none, there is nothing to lose. (Retention class, at the conjecture level of the core; no Page-curve or evaporation dynamics are claimed.)

(ii) MOND scale — a_0 grounded. MOND's standing complaint is that its acceleration a_0 is a *free measured constant*: the algorithm is untethered. Local-now supplies a tether, from Row-1 geometry alone (no temperature). A body's self-field reach c^2/a can cohere with the cosmic clock out to one full cycle, the horizon circumference $2\pi R_H = 2\pi c/H_0$. Dynamics therefore depart from Newton at $a_0 = c^2/2\pi R_H = cH_0/2\pi$ (Appendix B), where the 2π is the clock's radians-per-cycle rather than a thermal factor. The relevant horizon is the late de Sitter (future-event) horizon, the same one §2 requires for $w \to -1$. So a_0 is *constant* at $\sim cH_0$ today rather than tracking $H(z)$, which avoids tension with high-z disks. The Milgrom interpolation

$$\mu = \frac{a}{\sqrt{a^2 + a_0^2}}$$

then follows from a two-horizon quadrature (Rindler $\kappa_a = a$, de Sitter cH_0) [53, 54] *as an assignment, not a theorem.* The modified-*inertia* route that would distinguish DTF from MOND is *falsified* (stellar-stream apsides); orbits are MOND's modified-*gravity* (AQUAL) shadow, with no new galactic prediction. Appendix B carries the grade and the caveats. The chief one: cH_0, $cH_0/2\pi$ and $cH_0/6$ all sit inside the SPARC band (0.205 against 0.201 dex [55]), so the fit alone does not pin the coefficient.

(iii) Cosmological origin — time's emergence. The Big Bang is the *onset of a coherent now* (the Row-0→Row-1 transition at cosmic scale), the time-reverse of the core (i): coherence sets in at the Planck floor $N = 1$ (finite, not $a = 0$), so there is no initial singularity and no "before," and the horizon problem does not arise — uniformity is inherited from the timeless Row-0 floor, not established across a global slice. Whether the emergence also drives inflation is *left open* (it would need a transiently incomplete invariance, $\pi_u \neq 0$; O4).

Late times: the redshift is unchanged ($1 + z = a_{\rm obs}/a_{\rm emit}$; homogeneous $u(t)$ is gauge), and the one finite causal horizon $L = c/H_0$ fixes both dark scales,

$$\Omega_\Lambda = d^2 \sim O(1), \qquad a_0 \sim cH_0, \qquad \rho_\Lambda/\rho_P \sim (\ell_P/L)^2 \sim 10^{-122}.$$

All at conjecture level: the observed $\Omega_\Lambda = 0.68$ *would require* $d \approx 0.82$ (not forced), and $w = -1$ needs the future event horizon (Hsu–Li [18, 19]). Scripts: `DTF_cosmology_timeline`, `DTF_de_horizon`, `DTF_darksectors_unified`.

8 Limitations and open problems

These open problems are of a particular kind, and the kind follows from the choice this paper makes. Adding no equation to general relativity or quantum mechanics means the debts owed here are arguments rather than measurements; a framework that changed predictions would owe experiments instead, and should be judged by them. Several results above are supplied with motivated resolutions rather than theorems, and are listed here at that strength. On single-time statistics and on any complete-plane flux, DTF is *provably* identical to quantum mechanics (§6), so no experiment of that kind can discriminate; the framework's gravitational exposure is the black-hole remnant (§7),

and within quantum mechanics no channel survives at all (§6): the divergence-free freedom that could have carried one is not sourced by the tax. Two quantities that might be expected to serve as discriminators do not: monotonic mean curvature ($\dot{H} \leq 0$) requires a preferred slice this framework does not have, and the decaying-vorticity residual is unobservable by fifty orders of magnitude (§2).

O1. Outcome selection. Guidance $v = \nabla S/m$ plus an actual beable makes $|\psi|^2$ the typicality measure over the beable's unknown initial value; the residual — *why this initial configuration* — is the typicality floor every no-collapse theory shares, not a debt peculiar to DTF (§6).

O2. Black-hole core coefficient. The core *location* is coefficient-free ($N = L_{\rm curv}/\ell_P = 1$, Planck curvature, for any $p \in [\frac{1}{2}, 1]$); open is the $O(1)$ coefficient of $\delta u/u \sim (\ell_P/L)^p$ — the precise shell radius — needing a Row-0 foam propagator (§7).

O3. Does the projection generate the tensor structure? No-signalling follows from handle-local reads plus completeness (§6), and the row ordering explains why reads should be handle-local — but the two-handle tensor structure is assumed, not built. Whether the Row-0→Row-2 projection *generates* that structure or presupposes it decides whether no-signalling is derived or only motivated here.

O4. Transient inflation. If the Row-0→Row-1 emergence drives inflation (§7), it needs a transiently incomplete reparametrization invariance ($\pi_u \neq 0$); an action for that phase and any CMB observables (n_s, r) are unshown (§7).

O5. The driver of clock desync. The arrow follows from a spread σ_u in local rates (§2, App. G), but what *grows* σ_u from the Row-0 synced state is gravitational clumping — argued from the sign of the instability, not simulated. The entropy results are conditional on it.

Technical Appendix

A. Two polarisations and $\alpha_1 = \alpha_2 = 0$ (§5). Count in ADM variables [56]: the spatial metric γ_{ij} and conjugate π^{ij} give a 12-dimensional phase space per point, with four first-class constraints (one Hamiltonian, three momentum), each removing two dimensions:

$$N_{\rm dof} = \tfrac{1}{2}(12 - 2 \times 4) = 2 \qquad \text{(the transverse-traceless pair } h_{ij}^{TT}\text{)},$$

exactly the GR count, since the dynamics are Einstein's. The lapse and shift are multipliers; the lapse has $\pi_u \equiv 0$ (§2), so it contributes no scalar (breathing) mode and no vector clock-current. That absence forces the preferred-frame parameters to vanish — the gravitomagnetic g_{0i} from boosting the static solution and from solving for a moving source agree exactly:

$$\Delta g_{0i} = 0 \quad \Rightarrow \quad \alpha_1 = \alpha_2 = 0.$$

Scripts: `DTF_radiative_content_constructive.py`, `DTF_alpha1_explicit_1PN.py`.

B. The interpolation function and a_0 (§7), a Row-1 derivation. The MOND scale is derived here from Row-1 quantities only — causal-horizon geometry and the rate primitive — with *no* appeal to temperature, which is a Row-2 (statistical) object and may not enter a derivation of a Row-1 scale.

The scale. The primitive is a clock, and a clock is defined by cycles: one cycle is 2π of phase. The cosmic causal horizon has radius $R_H = c/H_0$, so one full cycle of the horizon-scale clock spans its

circumference $2\pi R_H$. A body's self-field reach is the Rindler length $L_a = c^2/a$. Coherence with the cosmic clock is lost once the reach exceeds one cosmic cycle, $L_a = 2\pi R_H$, giving

$$a_0 = \frac{c^2}{2\pi R_H} = \frac{cH_0}{2\pi} = 1.08 \times 10^{-10}\ \mathrm{m\,s^{-2}},$$

against Milgrom's measured 1.20×10^{-10} (a $\sim 10\%$ match under a $\sim 20\%$ systematic). The 2π is the fundamental clock's radians-per-cycle times the horizon circumference — both Row-1 — *not* a thermal 2π; a temperature route (Unruh $\propto a/2\pi$ against de Sitter $\propto H_0/2\pi$) instead *cancels* the 2π and gives $a_0 = cH_0$, which is why DTF does not take it.

The interpolation. A body's acceleration and the cosmic horizon define two *surface gravities* $\kappa_a = a$ and $\kappa_{\mathrm{dS}} = cH_0 = a_0$ — geometric, Row-1. They add in quadrature [53],

$$\kappa_{\mathrm{eff}} = \sqrt{a^2 + a_0^2},$$

and assigning the acceleration-attributable share $\mu = \kappa_a/\kappa_{\mathrm{eff}} = a/\sqrt{a^2 + a_0^2}$ reproduces Milgrom's form. This projection is an *assignment*, not a theorem, and it uses only surface gravities — geometry, not temperature.

Fit and caveats. At fixed $a_0 = cH_0/2\pi$ the 175-galaxy SPARC scatter is 0.205 dex, versus 0.201 at best-fit a_0 — a 2% degradation, so "zero free parameters" is defensible, though the fit alone cannot distinguish 2π from a nearby $O(1)$, and the one residual is firming that the reach saturates at *exactly* one cycle (not a half or two). Orbits precess as MOND and dark matter do (Orphan [57], Sagittarius [58]); the modified-*inertia* route that would have distinguished DTF is falsified, so DTF reproduces MOND's modified-*gravity* (AQUAL) shadow. This is distinct from thermodynamic entropic-gravity routes [27, 28], which pass through Row-2 temperature. Scripts: `DTF_mond_quadrature_zeta.py`, `DTF_sparc_fixed_a0.py`, `DTF_stream_apsidal.py`.

C. Schrödinger from the localization tax (§6). In polar form $\psi = \sqrt{\rho}\, e^{iS/\hbar}$ the Schrödinger equation

$$i\hbar\, \partial_t \psi = -\tfrac{\hbar^2}{2m}\nabla^2\psi + V\psi$$

separates *exactly* [35] into an imaginary part (continuity, conservation of the rate density) and a real part (the quantum Hamilton–Jacobi equation):

$$\partial_t \rho + \nabla\cdot\left(\rho\, \tfrac{\nabla S}{m}\right) = 0, \qquad \partial_t S + \frac{(\nabla S)^2}{2m} + V + Q = 0, \qquad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}.$$

The first two terms of the real part are §2's classical clock-action; the sole non-classical term is Q, and it is the Euler–Lagrange variation of the localization-tax (Fisher/von Weizsäcker) functional,

$$Q = \frac{\delta}{\delta\rho}\int \frac{\hbar^2}{8m}\frac{(\nabla\rho)^2}{\rho}\, d^3x \qquad \text{(Reginatto [36])},$$

carrying the same $\hbar$ the ledger names in $\lambda_C = \hbar/mc$. Both identities are verified symbolically (`DTF_localization_tax.py`). Born's measure is not a second postulate: the native current $j = \rho\nabla S/m$ makes $\rho = |\psi|^2$ the unique equivariant density, read out by all-or-nothing detection. *An operational transit speed.* Guidance is the current $v = \nabla S/m$ (Born-equivariant). In a forbidden region $\psi \sim e^{-\kappa x}$ is real, so $\nabla S \to 0$ (beable at rest), yet the tax is nonzero, giving an *operational* speed

$$Q = -\tfrac{\hbar^2}{2m}\kappa^2, \qquad v_{\mathrm{op}} = \sqrt{2|Q|/m} = \hbar\kappa/m,$$

which matches the 10^3–$10^4\,\mathrm{km\,s^{-1}}$ magnitudes of Sharoglazova *et al.* [44] at photonic mass (`DTF_evanescent_speed.p`). Two caveats: the $O(1)$ coefficient is not pinned by the data (as for a_0); and v_{op} is *not* a beable velocity — the challenge to Bohmian mechanics has been rebutted [46, 47], the authors maintaining their position [48].

D. Electromagnetism on the clock-phase (compatibility, §6). Only phase differences are physical, so the phase origin may be reset locally, $\psi \to e^{i\alpha(x)}\psi$; a connection restores covariance,

$$A_\mu \to A_\mu + \tfrac{\hbar c}{q}\partial_\mu\alpha, \qquad D_\mu = \partial_\mu - \tfrac{iq}{\hbar c}A_\mu.$$

Then $F_{\mu\nu}$ is gauge-invariant, the homogeneous Maxwell pair holds identically, varying $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$ gives the source pair, a mass term is forbidden (massless photon), and minimal coupling adds the Lorentz force to the §2 drive:

$$m\ddot{\mathbf{x}} = -mc^2\nabla u + q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$$

(`DTF_em_from_phase.py`). This is the standard $U(1)$ construction under the phase = clock-reading identification — a *consistency check*, not a derivation: DTF fixes neither α nor the charge assignments. Two points make it Row-native, not bolted on:

- **Local reset is not an extra postulate.** A global phase convention is the global reference §2 forgoes; with none, the origin is fixed only locally. A globally single-valued phase exists iff $F \equiv 0$, so denying a global referent is what admits a radiating $F \neq 0$ (`DTF_em_clockphase_forced.py`).
- **Clock-phase, not the frame.** Gauging the frame's internal $SO(3)$ instead gives three Planck-mass vectors and no photon (the tetrad VEV is a full-rank order parameter, `DTF_em_frame_gauge.py`); gravity (frame) and electromagnetism (frame-orthogonal $U(1)$) exhaust the long-range content from the triad VEV (`DTF_force_census.py`).

Charge quantisation would rest on the single-valuedness discussed in §6, so none is asserted.

E. Inputs and outputs. What goes in versus what comes out:

Inputs	**Outputs**
c, G, $\hbar$, H_0	GR weak-field limit; Kerr inherited
particle mass spectrum (ω_C values)	two tensor GW pols; $\alpha_1 = \alpha_2 = 0$
no structure added to GR (frame-neutral local rate)	inertia $= mc^2(\gamma - 1)$; equivalence as identity
localization-tax (Fisher) form; Born typicality	Schrödinger = clock-action + tax; $\lvert\psi\rvert^2$ measure
	$a_0 = cH_0/2\pi$ (coefficient, plausibility level)

F. The economy ledger and posit-counting. Stated as a ledger, the economy is a drop in the count of independent posits: things general relativity and quantum mechanics assume separately become one thing seen from different sides.

Posited separately in GR/QM	**Once the rate is the primitive**	**Grade**
Equivalence principle ($m_{\text{grav}} = m_{\text{inertial}}$)	one rate: never two masses to equate (§4)	dissolved
Gravity as spin-2 graviton exchange	no force-carrier; the propagating pair is space's radiation, not a mediator (§5)	reclassified
Preferred frame for realist QM	not needed: local rate + timeless Row 0 (§2,§6)	dissolved
Energy vs. frequency ($E = \hbar\omega$)	energy *is* the rate; definitional (§2)	identity
Gravity and inertia (two mechanisms)	one rate, read from two sides (§4,§5)	unified
Classical vs. quantum (two theories)	stationary vs. summed reading of one Φ (§2,§6)	unified
Born probability (separate axiom)	the rate-density itself; surprise isolated to O1 (§6)	clarified
Space (independent primitive)	the rate's radiative sector (§5)	reconceived
Guidance law chosen from a family (Deotto–Ghirardi non-uniqueness)	the current is conjugate to the phase in the action, not an independent postulate (§6)	dissolved
Past hypothesis (low-entropy start, posited)	$N = 1$ *is* Row 0; not assumable away (§2)	dissolved
Growth of the entropy (the arrow's rate)	ordinary mixing, inherited not derived; driver open (O5)	inherited

The ledger is an argument only if "one posit" has a criterion a skeptic cannot re-count away. We use one: *a posit is an independent input if the theory stays consistent and predictive when it is negated with the others held.* By that test the ledger is deliberately conservative — most rows grade below "dissolved": the equivalence principle *dissolves* (GR is consistent with $m_{\text{grav}} \neq m_{\text{inertial}}$, but with one ω_C the inequality is not expressible); $E = \hbar\omega$ is an *identity* (a unit conversion, not a removed posit); the preferred frame *dissolves* (rivals must *add* a negatable foliation, DTF has none to add); Born is only *clarified* (typicality survives negation, the axiom stands). The honest boundary: individuation is not fully observer-independent, so the claim is economy of *roles* played by one quantity — never a count of dimensionful constants, which stay inputs (Appendix E).

G. The entropy ledger (§2): one entropy, one count. Throughout this appendix the entropy is $S = \ln N$, the ordinary logarithm of a count of distinguishable states. No second quantity is introduced. What is derived is N — how many states the clock has resolved — and every number below is obtained without temperature; heat appears in exactly one paragraph, as an alternative way of reaching a large N.

One count, three readings. With N the accumulated distinguishable ticks,

$$E = N\hbar\omega, \qquad \Phi = 2\pi N, \qquad S = \ln N,$$

so $S = \ln(E/\hbar\omega)$: entropy and energy are not independent bookkeeping. At Row 0, $N = 1$ and $S = 0$.

The desync law, and what it does not do. Give the local rates a spread σ_u about the mean. Phases separate as $\phi_i = \omega u_i t$, so the occupied phase width — the tick count — is $N \simeq \omega\sigma_u t$ and $S = \ln N = \ln(\omega\sigma_u t)$. Numerically ($4 \times 10^5$ clocks, 4096 phase cells, unsaturated window): the fit $S = a \ln t + b$ gives $a = 0.982$ against the predicted 1; doubling σ_u at fixed t shifts S by 0.6936 and doubling ω by 0.6924, against $\ln 2 = 0.6931$ — both to better than 0.1%. Two disclaimers, since this is the section's most quotable number. First, the input was $N \propto \omega\sigma_u t$, so the fit verifies that the code implements the definition and that ω, σ_u act multiplicatively inside N; it is not independent evidence that entropy grows. Second, the simulated quantity is $-\sum p \ln p$, which sits 0.70–0.79 nats below ln(occupied cells)

for a Gaussian spread — an additive offset, not a change in the law — so the honest statement is $S = \ln N + O(1)$, with $\ln N$ the equal-weight ceiling. A discrete, commensurate set of rates instead recurs (Poincaré; S oscillates in $[0, 1.79]$): the law is reversible, the monotone arrow needs many degrees of freedom plus the $N = 1$ start, and DTF's contribution is the start, not the mixing.

Purity is preserved. In a transverse-field Ising quench the global state stays pure ($S_{\text{global}} = 0$ to machine precision at all times) while a subsystem's entanglement entropy grows linearly in the tick count (slope 0.663) before saturating at ln(dim). The Row-1 arrow is therefore the *redistribution* of a conserved Row-0 correlation budget, which is why nothing is destroyed at a black-hole core (§7).

The rate bound. Margolus–Levitin [59] caps the tick rate at $dN/d\tau \leq 2E/\pi\hbar$; applied to the observable universe this returns Lloyd's $N \sim 10^{120}$ ticks [60]. The arrow's maximum speed *is* the energy — the same identity as $E = N\hbar\omega$ differentiated.

Reaching a large count by heating instead. Put the oscillator in a bath, so occupation rather than desync resolves the states. With the zero-point half-tick included the count is $N_{\text{th}} = \frac{1}{2} + 1/(e^{\hbar\omega/k_BT} - 1)$, and the large-$N$ form of the same $S = \ln N$ is $\ln(eN_{\text{th}})$. Against the exact oscillator entropy $S = (1+n)\ln(1+n) - n\ln n$ with $n = 1/(e^{\hbar\omega/k_BT} - 1)$:

$k_BT/\hbar\omega$	N_{th}	exact S	$\ln(eN_{\text{th}})$
0.5	0.657	0.458	0.579
1	1.082	1.041	1.079
5	5.017	2.611	2.613
50	50.00	4.912	4.912
500	500.0	7.215	7.215

The agreement is *asymptotic*: four digits once $N_{\text{th}} \gtrsim 5$, but a 26% error at $N_{\text{th}} < 1$, where a count-of-states reading is not valid anyway. (Dropping the zero-point half-tick makes $\ln(eN)$ negative at low T, which is the diagnostic that the half belongs in the count.) The point is not numerical precision but that nothing about the *entropy* changed — only what resolved the states. That ordering is not ours to choose: temperature is *defined* from entropy ($1/T = \partial S/\partial E$), and Landauer's $k_BT\ln 2$ likewise separates a count $\ln 2$ from a price k_BT [23, 24]. A count without a bath is routine physics — it survives at $T \to 0$ [25], it is what is measured in a globally pure state [22], and the Kolmogorov–Sinai rate counts per unit time with no temperature in it [26]. DTF adds no entropy; it identifies N with the clock.

Which share dominates. Measured budgets [30] put the black-hole share of the observable universe at $3.1 \times 10^{104} k_B$ against the microwave background's $\sim 10^{88} k_B$, with the cosmic event horizon at $2.6 \times 10^{122} k_B$; our coarse estimate reproduces these to an order of magnitude. Radiation holds a negligible fraction of N, and the dominant share — carried by clumping, i.e. by departures of u from uniformity — is near its minimum when u is uniform. A hot but smooth start is therefore a *small* N, which is Penrose's point [29] restated as a question of shares rather than an added postulate. Honest limit: how to count the clumping share has no agreed formula, and black-hole entropy is doing duty as the proxy. Scripts: `DTF_entropy_clockcycle.py`, `DTF_native_arrow.py`, `DTF_entropy_is_entanglement.py`, `DTF_arrow_grav_thermal.py`.

Code availability

All results marked *verified* are reproduced by short Python scripts (NumPy/SciPy/SymPy) at `https://github.com/billd-dtf/Distorted-Time-Framework` (all names carry the `DTF_` prefix):

- **§2:** `which_reading_is_DTF`, `phase_unification`, `o6_congruence`, `localnow_integrability`, `native_arrow`, `entropy_clockcycle`, `entropy_is_entanglement`, `arrow_grav_thermal`, `vorticity_residual`
- **§4:** `kerr_deser_closure`
- **§5:** `radiative_content_constructive`, `alpha1_explicit_1PN`
- **§6:** `localization_tax`, `evanescent_speed`, `born_relaxation_2d`, `o8_tsirelson`, `o9_fisher_unique`, `o1_wallstrom`, `o2_deotto_ghirardi`, `winding_dynamics`, `o2_amplitude_channel`, `o2_arrival_time`, `o2_tax_sources_c` `no_signalling_projection`
- **§7:** `singularity_deemergence`, `LTB_collapse`, `singularity_observable`, `mond_quadrature_zeta`, `sparc_fixed_a0`, `stream_apsidal`, `cosmology_timeline`, `de_horizon`, `darksectors_unified`, `localnow_eventhorizon`
- **App. D:** `em_from_phase`, `em_clockphase_forced`, `force_census`, `em_frame_gauge`

Note on methods

The author specified the physical targets, claims, and their epistemic grades; a large language model (Anthropic Claude) was used as a collaborator for derivations, symbolic and numerical checks, and drafting. All results flagged *verified* are reproduced by the scripts above, and the author is responsible for the content.

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